URL Shortener: Stage 1 Final Review

Project	`com.malinda:url-shortener` (`E:\Projects\URL Shortener\url-shortener`)
Branch / HEAD	`dev` @ `cd61f9d` "Implement URL validation, error handling, and refactor URL creation logic" (2026-11-08 23:44 +0530)
Review date	2026-11-08/09 (tests run 2026-11-08 23:56 +0530)
Reviewer	Claude Code (Claude Opus 5.5), at the author's request
Compared against	The 5 November review item list supplied by the author (no earlier review file exists in `docs/reviews/`)

1. Summary

Every invalid input in the contract returns 400 with no stack trace. Duplicate submissions return 200 with the existing code. Redirects are 302 and keep percent-encoding intact. **Stage 1 is not complete.** `GET /{unknown}` returns **500 with a stack trace** instead of 404, so `UrlShortenerAcceptanceTest.unknownCodeReturns404` fails (14 of 15 tests pass). `shortUrl` is still wrong when `app.base-url` has no trailing slash (`http://localhost:18082EOQGkA`). I also observed that non-ASCII URLs come back percent-encoded in `Location`. Of the 5 November items, 3 are fixed, 2 are partly fixed and 13 are still open (see section 11). No code under `src/main`, `pom.xml` or config was changed in this session.

2. Stage 1 exit checklist

Criterion	Result	Evidence
All tests pass	**FAIL**	15 run, 14 pass, 1 error: `unknownCodeReturns404` (section 6)
Every contract case verified by curl	**FAIL**	All contract cases were exercised. Every case behaved as the contract says except "unknown code -> 404", which returned 500 (section 7, check 4)
No 5xx for any client error	**FAIL**	Unknown code -> `HTTP/1.1 500` (section 7, check 4)
No stack traces for 4xx cases	**PASS**	All 10 required 4xx cases logged nothing, or one `WARN` line with no stack trace. The only stack trace came from the 500 (unknown code)
Parked items left alone	**PASS**	Collision handling, the check-then-act race, the linear scan and the static generator are unchanged and left for later stages (section 10)

3. Environment

Item	Value	Source
Java	21.0.7 (Oracle HotSpot, build 21.0.7+8-LTS-245)	`java -version`, `mvnw -v`
Java target	21 (`<java.version>21</java.version>`; javac ran with `release 21`)	`pom.xml`, build log
Spring Boot	4.1.2 (`spring-boot-starter-parent`)	`pom.xml` line 8, startup banner `:: Spring Boot :: (v4.1.2)`
Build tool	Maven 3.10.0 via Maven Wrapper 3.3.4 (`mvnw`, `distributionType=only-script`)	`.mvn/wrapper/maven-wrapper.properties`, `mvnw -v`
Dependencies	`spring-boot-starter-webmvc`, `spring-boot-starter-webmvc-test` (test)	`pom.xml`

Item	Value	Source
Surefire	3.5.6	build log
OS	Windows 11 (10.0.26200), curl 8.14.1	

`pom.xml` changes since the 5 November review: none. `git log -- pom.xml` shows that the last commit touching it is `607f837` (2026-11-02), which added the `maven-compiler-plugin` block with `--enable-preview`. Neither commit after 5 November (`865430c` 2026-11-06, `cd61f9d` 2026-11-08) changed `pom.xml`.

4. File tree of `src/`

```
src/
    main/
        java/
            com/
                malinda/
                    url_shortener/
                        UrlShortenerApplication.java
                        controller/
                            UrlController.java
                        dto/
                            ApiErrorResponse.java
                            CreateUrlRequest.java
                            CreateUrlResponse.java
                            CreateUrlResult.java
                        exceptions/
                            GlobalExceptionHandler.java
                            InvalidUrlException.java
                        service/
                            UrlService.java
                            IMPL/
                                UrlServiceIMPL.java
                        util/
                            ShortCodeGenerator.java
                            UrlValidator.java
        resources/
            application.yaml
            static/
            templates/
    test/
        java/
            com/
                malinda/
                    url_shortener/
                        UrlShortenerAcceptanceTest.java
                        UrlShortenerApplicationTests.java
                        UrlShortenerErrorCasesTest.java
```

`resources/static/` and `resources/templates/` are empty directories. `UrlShortenerErrorCasesTest.java` was created in this session (Step 2). Everything else is the author's.

5. Full contents of every file under `src/main`

These are the verbatim contents at HEAD `cd61f9d`. The working tree has no changes under `src/main` (`git status --short` lists only the new test file).

5.1 `src/main/java/com/malinda/url_shortener/UrlShortenerApplication.java`

```
1  package com.malinda.url_shortener;
2
3  import org.springframework.boot.SpringApplication;
4  import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6  @SpringBootApplication
7  public class UrlShortenerApplication {
```

```
 8
 9     public static void main(String[] args) {
10             SpringApplication.run(UrlShortenerApplication.class, args);
11     }
12
13 }
```

5.2 `src/main/java/com/malinda/url_shortener/controller/UrlController.java`

```
 1  package com.malinda.url_shortener.controller;
 2
 3  import com.malinda.url_shortener.dto.CreateUrlRequest;
 4  import com.malinda.url_shortener.dto.CreateUrlResponse;
 5  import com.malinda.url_shortener.dto.CreateUrlResult;
 6  import com.malinda.url_shortener.service.UrlService;
 7  import org.springframework.http.HttpStatus;
 8  import org.springframework.http.ResponseEntity;
 9  import org.springframework.web.bind.annotation.*;
10
11  import java.net.URI;
12
13  @RestController
14  public class UrlController {
15
16      private final UrlService urlService;
17
18      public UrlController(UrlService urlService) {
19          this.urlService = urlService;
20      }
21
22      @PostMapping("/api/urls")
23      public ResponseEntity<CreateUrlResponse> createShortUrl(@RequestBody CreateUrlRequest request) {
24          CreateUrlResult response = urlService.createShortUrl(request.url());
25
26          if (response.created()){
27              return ResponseEntity
28                      .status(HttpStatus.CREATED)
29                      .body(response.response());
30          } else {
31              return ResponseEntity
32                      .status(HttpStatus.OK)
33                      .body(response.response());
34          }
35      }
36
37      @GetMapping("/{code}")
38      public  ResponseEntity<Void> redirectToOriginalUrl(@PathVariable String code) {
39          String response = urlService.redirectToOriginalUrl(code);
40          return  ResponseEntity
41                  .status(HttpStatus.FOUND)
42                  .location(URI.create(response))
43                  .build();
44      }
45  }
```

5.3 `src/main/java/com/malinda/url_shortener/dto/ApiErrorResponse.java`

```
1  package com.malinda.url_shortener.dto;
2
3  public record ApiErrorResponse(String message) {
4  }
```

5.4 `src/main/java/com/malinda/url_shortener/dto/CreateUrlRequest.java`

```
1  package com.malinda.url_shortener.dto;
2
3  public record CreateUrlRequest(String url) {}
```

5.5 `src/main/java/com/malinda/url_shortener/dto/CreateUrlResponse.java`

```java
package com.malinda.url_shortener.dto;

public record CreateUrlResponse(String code, String shortUrl) {}
```

5.6 `src/main/java/com/malinda/url_shortener/dto/CreateUrlResult.java`

```java
package com.malinda.url_shortener.dto;

public record CreateUrlResult(CreateUrlResponse response, boolean created) {
}
```

5.7 `src/main/java/com/malinda/url_shortener/exceptions/GlobalExceptionHandler.java`

```java
package com.malinda.url_shortener.exceptions;

import com.malinda.url_shortener.dto.ApiErrorResponse;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.RestControllerAdvice;

@RestControllerAdvice
public class GlobalExceptionHandler {

    @ExceptionHandler(InvalidUrlException.class)
    public ResponseEntity<ApiErrorResponse> handleInvalidUrlException(InvalidUrlException e) {
        return ResponseEntity
                .status(HttpStatus.BAD_REQUEST)
                .body(new ApiErrorResponse(e.getMessage()));
    }
}
```

5.8 `src/main/java/com/malinda/url_shortener/exceptions/InvalidUrlException.java`

```java
package com.malinda.url_shortener.exceptions;

public class InvalidUrlException extends RuntimeException{
    public InvalidUrlException(String message) {
        super(message);
    }
}
```

5.9 `src/main/java/com/malinda/url_shortener/service/UrlService.java`

```java
package com.malinda.url_shortener.service;

import com.malinda.url_shortener.dto.CreateUrlResponse;
import com.malinda.url_shortener.dto.CreateUrlResult;

public interface UrlService {
    CreateUrlResult createShortUrl(String longUrl);

    String redirectToOriginalUrl(String code);
}
```

5.10 `src/main/java/com/malinda/url_shortener/service/IMPL/UrlServiceIMPL.java`

```java
package com.malinda.url_shortener.service.IMPL;

import com.malinda.url_shortener.dto.CreateUrlResponse;
import com.malinda.url_shortener.dto.CreateUrlResult;
import com.malinda.url_shortener.service.UrlService;
import com.malinda.url_shortener.util.ShortCodeGenerator;
```

```java
import com.malinda.url_shortener.util.UrlValidator;
import org.springframework.beans.factory.annotation.Value;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.stereotype.Service;

import java.util.Map;
import java.util.concurrent.ConcurrentHashMap;

@Service
public class UrlServiceIMPL implements UrlService {

    private final Map<String, String> urlMappings = new ConcurrentHashMap<>();

    @Value("${app.base-url}")
    private String baseUrl;

    @Override
    public CreateUrlResult createShortUrl(String longUrl) {
        UrlValidator.validateUrl(longUrl);


        for (Map.Entry<String, String> entry : urlMappings.entrySet()) {

            if (entry.getValue().equals(longUrl)) {
                String existingCode = entry.getKey();

                return new CreateUrlResult(
                        new CreateUrlResponse(
                                existingCode,
                                baseUrl + existingCode),
                        false
                );
            }
        }

        String code = ShortCodeGenerator.generateShortCode();
        urlMappings.put(code, longUrl);
        return new CreateUrlResult(
                new CreateUrlResponse(
                        code,
                        baseUrl + code),
                true
        );
    }

    @Override
    public String redirectToOriginalUrl(String code) {
        if (code.isEmpty() || !urlMappings.containsKey(code)) {
            return "Invalid code";
        } else {
            return urlMappings.get(code);
        }
    }
}
```

5.11 `src/main/java/com/malinda/url_shortener/util/ShortCodeGenerator.java`

```java
package com.malinda.url_shortener.util;

import java.security.SecureRandom;

public class ShortCodeGenerator {
    private static final String CHARACTERS =
            "0123456789abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ";

    private static final SecureRandom secureRandom = new SecureRandom();

    public static String generateShortCode() {
        StringBuilder shortCode = new StringBuilder();

        for (int i = 0; i < 6; i++) {
            int randomIndex = secureRandom.nextInt(CHARACTERS.length());
            shortCode.append(CHARACTERS.charAt(randomIndex));
```

```
 17          }
 18
 19          return shortCode.toString();
 20      }
 21  }
```

5.12 `src/main/java/com/malinda/url_shortener/util/UrlValidator.java`

```
  1  package com.malinda.url_shortener.util;
  2
  3  import com.malinda.url_shortener.exceptions.InvalidUrlException;
  4
  5  import java.net.URI;
  6
  7  public class UrlValidator {
  8      public static void validateUrl(String longUrl) {
  9          if (longUrl == null || longUrl.isBlank()) {
 10              throw new InvalidUrlException("URL cannot be empty");
 11          }
 12
 13          try {
 14              URI uri = URI.create(longUrl);
 15
 16              String schema = uri.getScheme();
 17
 18              if (schema == null || (!schema.equalsIgnoreCase("http") && !schema.equalsIgnoreCase("https")) ||
uri.getHost() == null) {
 19                  throw new InvalidUrlException("Invalid URL format");
 20              }
 21          } catch (IllegalArgumentException e) {
 22              throw new InvalidUrlException("Malformed URL");
 23          }
 24      }
 25  }
```

5.13 `src/main/resources/application.yaml`

```
  1  spring:
  2    application:
  3      name: url-shortener
  4
  5  server:
  6    port: 8081
  7
  8  app:
  9    base-url: http://localhost:8081/
```

Line numbers are added for reference and are not part of the files. The findings in section 8 cite these numbers.

6. Test files, commands and results

6.1 Test files

`UrlShortenerAcceptanceTest.java` (existing, author's, unchanged since `234e362` 2026-11-02):

```
package com.malinda.url_shortener;

import com.jayway.jsonpath.JsonPath;
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.webmvc.test.autoconfigure.AutoConfigureMockMvc;
import org.springframework.http.MediaType;
import org.springframework.test.web.servlet.MockMvc;
import org.springframework.test.web.servlet.MvcResult;

import static org.assertj.core.api.Assertions.assertThat;
```

```
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.post;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.header;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;

/**
 * Black-box acceptance test for the Stage 1 API contract:
 *   POST /api/urls {"url": "..."} -> 201 {"code": "...", "shortUrl": "<base>/<code>"}
 *   GET  /{code}                  -> 3xx, Location: original url | 404 if unknown
 * Exercises the app only through HTTP (MockMvc does not follow redirects).
 */
@SpringBootTest
@AutoConfigureMockMvc
class UrlShortenerAcceptanceTest {

    private static final String TRACKING_URL =
            "https://shop.example.com/products/12345?utm_source=sms&utm_campaign=launch";
    private static final String ENCODED_URL = "https://example.com/search?q=hello%20world";

    @Autowired
    private MockMvc mockMvc;

    /** POSTs a URL, asserts the creation contract, and returns the code. */
    private String shorten(String longUrl) throws Exception {
        String body = "{\"url\": " + jsonString(longUrl) + "}";
        MvcResult result = mockMvc.perform(post("/api/urls")
                        .contentType(MediaType.APPLICATION_JSON)
                        .content(body))
                .andExpect(status().isCreated())
                .andReturn();

        String json = result.getResponse().getContentAsString();
        String code = JsonPath.read(json, "$.code");
        String shortUrl = JsonPath.read(json, "$.shortUrl");

        assertThat(code).isNotBlank();
        assertThat(shortUrl).endsWith("/" + code);
        return code;
    }

    private static String jsonString(String s) {
        return "\"" + s.replace("\\", "\\\\").replace("\"", "\\\"") + "\"";
    }

    @Test
    void createReturns201WithCodeAndShortUrl() throws Exception {
        shorten("https://example.com/some/page");
    }

    @Test
    void shortCodeRedirectsToExactOriginalUrl() throws Exception {
        String code = shorten(TRACKING_URL);

        mockMvc.perform(get("/" + code))
                .andExpect(status().is3xxRedirection())
                .andExpect(header().string("Location", TRACKING_URL));
    }

    @Test
    void percentEncodedUrlIsNotDoubleEncoded() throws Exception {
        String code = shorten(ENCODED_URL);

        mockMvc.perform(get("/" + code))
                .andExpect(status().is3xxRedirection())
                .andExpect(header().string("Location", ENCODED_URL));
    }

    @Test
    void differentUrlsGetDifferentCodesAndRedirectToTheirOwnOriginal() throws Exception {
        String urlA = "https://a.example.com/first";
        String urlB = "https://b.example.com/second";

        String codeA = shorten(urlA);
        String codeB = shorten(urlB);

        assertThat(codeA).isNotEqualTo(codeB);
```

```
        mockMvc.perform(get("/" + codeA))
                .andExpect(status().is3xxRedirection())
                .andExpect(header().string("Location", urlA));
        mockMvc.perform(get("/" + codeB))
                .andExpect(status().is3xxRedirection())
                .andExpect(header().string("Location", urlB));
    }

    @Test
    void unknownCodeReturns404() throws Exception {
        mockMvc.perform(get("/neverCreatedCode123"))
                .andExpect(status().isNotFound());
    }
}
```

`UrlShortenerApplicationTests.java` (existing, generated by Spring Initializr):

```
package com.malinda.url_shortener;

import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;

@SpringBootTest
class UrlShortenerApplicationTests {

        @Test
        void contextLoads() {
        }

}
```

`UrlShortenerErrorCasesTest.java` (**created in this session**, because it did not exist; tests over HTTP only, with hosts under `errcases.example.com`):

```
package com.malinda.url_shortener;

import com.jayway.jsonpath.JsonPath;
import org.junit.jupiter.api.Test;
import org.junit.jupiter.params.ParameterizedTest;
import org.junit.jupiter.params.provider.ValueSource;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.webmvc.test.autoconfigure.AutoConfigureMockMvc;
import org.springframework.http.MediaType;
import org.springframework.test.web.servlet.MockMvc;
import org.springframework.test.web.servlet.MvcResult;

import static org.assertj.core.api.Assertions.assertThat;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.post;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;

/**
 * Black-box error-case and duplicate tests for the Stage 1 API contract:
 *   POST /api/urls new URL           -> 201 {"code", "shortUrl"}
 *   POST /api/urls already shortened -> 200 {"code", "shortUrl"} with the existing code
 *   POST /api/urls missing / blank / malformed / not http(s) -> 400
 * HTTP only (MockMvc). All test classes share one in-memory store, so every URL here
 * uses a host under errcases.example.com that appears in no other test.
 */
@SpringBootTest
@AutoConfigureMockMvc
class UrlShortenerErrorCasesTest {

    @Autowired
    private MockMvc mockMvc;

    private MvcResult postBody(String body) throws Exception {
        return mockMvc.perform(post("/api/urls")
                        .contentType(MediaType.APPLICATION_JSON)
                        .content(body))
                .andReturn();
```

```
    }

    @Test
    void sameUrlTwiceReturns201ThenExisting200WithSameCodeAndShortUrl() throws Exception {
        String body = "{\"url\": \"https://dup.errcases.example.com/page?x=1\"}";

        MvcResult first = mockMvc.perform(post("/api/urls")
                        .contentType(MediaType.APPLICATION_JSON)
                        .content(body))
                .andExpect(status().isCreated())
                .andReturn();
        MvcResult second = mockMvc.perform(post("/api/urls")
                        .contentType(MediaType.APPLICATION_JSON)
                        .content(body))
                .andExpect(status().isOk())
                .andReturn();

        String firstJson = first.getResponse().getContentAsString();
        String secondJson = second.getResponse().getContentAsString();
        String firstCode = JsonPath.read(firstJson, "$.code");
        String firstShortUrl = JsonPath.read(firstJson, "$.shortUrl");

        assertThat(firstCode).isNotBlank();
        assertThat((String) JsonPath.read(secondJson, "$.code")).isEqualTo(firstCode);
        assertThat((String) JsonPath.read(secondJson, "$.shortUrl")).isEqualTo(firstShortUrl);
    }

    @ParameterizedTest(name = "[{index}] {0} -> 400")
    @ValueSource(strings = {
            "{}",
            "{\"url\": \"\"}",
            "{\"url\": \"   \"}",
            "{\"url\": \"not-a-url\"}",
            "{\"url\": \"https://\"}",
            "{\"url\": \"javascript:alert(1)\"}",
            "{\"url\": \"ftp://errcases.example.com/file\"}",
            "{\"url\": \"https://errcases.example.com/a b\"}"
    })
    void invalidInputReturns400(String body) throws Exception {
        assertThat(postBody(body).getResponse().getStatus()).isEqualTo(400);
    }
}
```

6.2 Command

```
cd url-shortener
./mvnw -B test
```

Result: `Tests run: 15, Failures: 0, Errors: 1, Skipped: 0`, `BUILD FAILURE` (finished 2026-11-08T23:56:12+05:30, 21.3 s).

6.3 Per-test results (from `target/surefire-reports/TEST-*.xml`)

#	Class	Test	Result
1	UrlShortenerAcceptanceTest	createReturns201WithCodeAndShortUrl	PASS
2	UrlShortenerAcceptanceTest	shortCodeRedirectsToExactOriginalUrl	PASS
3	UrlShortenerAcceptanceTest	percentEncodedUrlIsNotDoubleEncoded	PASS
4	UrlShortenerAcceptanceTest	differentUrlsGetDifferentCodesAndRedirectToTheirOwnOriginal	PASS
5	UrlShortenerAcceptanceTest	unknownCodeReturns404	**ERROR**
6	UrlShortenerApplicationTests	contextLoads	PASS
7	UrlShortenerErrorCasesTest	sameUrlTwiceReturns201ThenExisting200WithSameCodeAndShortUrl	PASS
8	UrlShortenerErrorCasesTest	invalidInputReturns400 [1] `{}`	PASS
9	UrlShortenerErrorCasesTest	invalidInputReturns400 [2] `{"url": ""}`	PASS

#	Class	Test	Result
10	UrlShortenerErrorCasesTest	invalidInputReturns400 [3] `{"url": "   "}`	PASS
11	UrlShortenerErrorCasesTest	invalidInputReturns400 [4] `{"url": "not-a-url"}`	PASS
12	UrlShortenerErrorCasesTest	invalidInputReturns400 [5] `{"url": "https://"}`	PASS
13	UrlShortenerErrorCasesTest	invalidInputReturns400 [6] `{"url": "javascript:alert(1)"}`	PASS
14	UrlShortenerErrorCasesTest	invalidInputReturns400 [7] `{"url": "ftp://errcases.example.com/file"}`	PASS
15	UrlShortenerErrorCasesTest	invalidInputReturns400 [8] `{"url": "https://errcases.example.com/a b"}`	PASS

Failure detail for #5 (excerpt from the build log):

```
[ERROR] com.malinda.url_shortener.UrlShortenerAcceptanceTest.unknownCodeReturns404 -- Time elapsed: 0.017 s <<<
ERROR!
jakarta.servlet.ServletException: Request processing failed: java.lang.IllegalArgumentException: Illegal character in
path at index 7: Invalid code
        at org.springframework.web.servlet.FrameworkServlet.processRequest(FrameworkServlet.java:1008)
        ...
        at
com.malinda.url_shortener.UrlShortenerAcceptanceTest.unknownCodeReturns404(UrlShortenerAcceptanceTest.java:100)
Caused by: java.lang.IllegalArgumentException: Illegal character in path at index 7: Invalid code
        at java.base/java.net.URI.create(URI.java:932)
        at com.malinda.url_shortener.controller.UrlController.redirectToOriginalUrl(UrlController.java:42)
```

6.4 Notes on the existing tests (none edited)

- No existing test contradicts the contract. `unknownCodeReturns404` is correct, and it fails because of the application.
- Several acceptance tests use `status().is3xxRedirection()`, which is looser than the contract's 302: a 301 or 307 would also pass. The class Javadoc also says "3xx". This is a gap, not a wrong assertion.
- `shorten()` checks only that `shortUrl` ends with `"/" + code`. It does not check the base-URL prefix, so the missing-slash defect (finding C2) would not be caught. The test profile uses the yaml value, which has a trailing slash, so the defect is hidden anyway.
- The build log shows the JVM warnings "A Java agent has been loaded dynamically (byte-buddy-agent)" and "Sharing is only supported for boot loader classes". These are Mockito/JDK 21 notices and do not affect the results.

7. curl transcripts

Method: I packaged the jar with `./mvnw -B -q -DskipTests package` and ran it in the background with `java -jar target/url-shortener-0.0.1-SNAPSHOT.jar --server.port=<port> --app.base-url=<base>`. I ran the curl checks with `curl -s -i`, then stopped the process. Afterwards `netstat` showed no listeners on ports 18081 to 18083. The `Date`, `Connection`, `Keep-Alive`, `Transfer-Encoding` and `Vary` headers are trimmed. After each request, the `[server log: ...]` line reports what the app log gained during that request.

7.1 Checks 1 to 5 (port 18081, `--app.base-url=http://localhost:18081/`)

```
== App started: port 18081, --app.base-url=http://localhost:18081/ (pid 2150)

### 1a. Create a new link
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d
'{"url":"https://curl.example.com/articles/42?ref=manual"}'
HTTP/1.1 201
Content-Type: application/json

{"code":"MTbmNv","shortUrl":"http://localhost:18081/MTbmNv"}

[server log: no new lines; no stack trace]
```

```
### 1b. Follow it (code MTbmNv)
$ curl -i http://localhost:18081/MTbmNv
HTTP/1.1 302
Location: https://curl.example.com/articles/42?ref=manual
Content-Length: 0


[server log: no new lines; no stack trace]

### 2. Submit the same URL again (expect 200, code MTbmNv)
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d
'{"url":"https://curl.example.com/articles/42?ref=manual"}'
HTTP/1.1 200
Content-Type: application/json

{"code":"MTbmNv","shortUrl":"http://localhost:18081/MTbmNv"}

[server log: no new lines; no stack trace]

### 3a. Create the percent-encoded URL
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d
'{"url":"https://example.com/search?q=hello%20world"}'
HTTP/1.1 201
Content-Type: application/json

{"code":"8ptZwj","shortUrl":"http://localhost:18081/8ptZwj"}

[server log: no new lines; no stack trace]

### 3b. Follow it (code 8ptZwj)
$ curl -i http://localhost:18081/8ptZwj
HTTP/1.1 302
Location: https://example.com/search?q=hello%20world
Content-Length: 0


[server log: no new lines; no stack trace]

### 4. Unknown code
$ curl -i http://localhost:18081/zzzzzz
HTTP/1.1 500
Content-Type: application/json

{"timestamp":"2026-11-08T18:28:54.255Z","status":500,"error":"Internal Server Error","path":"/zzzzzz"}

[server log: STACK TRACE present (50 'at' frames)]
2026-11-08T23:58:54.250+05:30 ERROR 17700 --- [url-shortener] [io-18081-exec-4] o.a.c.c.C.[.[.[/].[dispatcherServlet]
: Servlet.service() for servlet [dispatcherServlet] in context with path [] threw exception [Request processing
failed: java.lang.IllegalArgumentException: Illegal character in path at index 7: Invalid code] with root cause
java.net.URISyntaxException: Illegal character in path at index 7: Invalid code

### 5a. Empty body (no content)
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json'
HTTP/1.1 400
Content-Type: application/json

{"timestamp":"2026-11-08T18:28:54.866Z","status":400,"error":"Bad Request","path":"/api/urls"}

[server log: no stack trace; single-line entry:]
2026-11-08T23:58:54.865+05:30  WARN 17700 --- [url-shortener] [io-18081-exec-5]
.w.s.m.s.DefaultHandlerExceptionResolver : Resolved
[org.springframework.http.converter.HttpMessageNotReadableException: Required request body is missing: public
org.springframework.http.ResponseEntity<com.malinda.url_shortener.dto.CreateUrlResponse>
com.malinda.url_shortener.controller.UrlController.createShortUrl(com.malinda.url_shortener.dto.CreateUrlRequest)]

### 5b. {}
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d '{}'
HTTP/1.1 400
Content-Type: application/json

{"message":"URL cannot be empty"}

[server log: no new lines; no stack trace]

### 5c. empty string
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d '{"url": ""}'
```

```
HTTP/1.1 400
Content-Type: application/json

{"message":"URL cannot be empty"}

[server log: no new lines; no stack trace]

### 5d. blank
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d '{"url": "   "}'
HTTP/1.1 400
Content-Type: application/json

{"message":"URL cannot be empty"}

[server log: no new lines; no stack trace]

### 5e. not-a-url
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d '{"url": "not-a-url"}'
HTTP/1.1 400
Content-Type: application/json

{"message":"Invalid URL format"}

[server log: no new lines; no stack trace]

### 5f. https:// (no host)
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d '{"url": "https://"}'
HTTP/1.1 400
Content-Type: application/json

{"message":"Malformed URL"}

[server log: no new lines; no stack trace]

### 5g. javascript scheme
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d '{"url":
"javascript:alert(1)"}'
HTTP/1.1 400
Content-Type: application/json

{"message":"Invalid URL format"}

[server log: no new lines; no stack trace]

### 5h. ftp scheme
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d '{"url":
"ftp://errcases.example.com/file"}'
HTTP/1.1 400
Content-Type: application/json

{"message":"Invalid URL format"}

[server log: no new lines; no stack trace]

### 5i. space in path
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d '{"url":
"https://errcases.example.com/a b"}'
HTTP/1.1 400
Content-Type: application/json

{"message":"Malformed URL"}

[server log: no new lines; no stack trace]

== Additional probes (outside the required list)

### X1. malformed JSON
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d '{"url":'
HTTP/1.1 400
Content-Type: application/json

{"timestamp":"2026-11-08T18:28:59.713Z","status":400,"error":"Bad Request","path":"/api/urls"}

[server log: no stack trace; single-line entry:]
2026-11-08T23:58:59.712+05:30  WARN 17700 --- [url-shortener] [io-18081-exec-4]
.w.s.m.s.DefaultHandlerExceptionResolver : Resolved
```

```
[org.springframework.http.converter.HttpMessageNotReadableException: JSON parse error: Unexpected end-of-input
within/between Object entries]

### X2. non-ASCII path, sent as a UTF-8 body file containing: {"url":"https://probe.example.com/café"}
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' --data-binary @body-cafe.json
HTTP/1.1 201
Content-Type: application/json

{"code":"Yesu2F","shortUrl":"http://localhost:18081/Yesu2F"}

[server log: no new lines; no stack trace]

### X2b. Follow it (code Yesu2F); stored URL is https://probe.example.com/café
$ curl -i http://localhost:18081/Yesu2F
HTTP/1.1 302
Location: https://probe.example.com/caf%C3%A9
Content-Length: 0


[server log: no new lines; no stack trace]

### X3. trailing-slash variant of an existing URL
$ curl -i -X POST http://localhost:18081/api/urls -H 'Content-Type: application/json' -d
'{"url":"https://curl.example.com/articles/42/?ref=manual"}'
HTTP/1.1 201
Content-Type: application/json

{"code":"Ow10Y7","shortUrl":"http://localhost:18081/Ow10Y7"}

[server log: no new lines; no stack trace]

== App stopped
```

Server-log excerpt for check 4 (unknown code), showing the stack trace:

```
2026-11-08T23:58:54.250+05:30 ERROR 17700 --- [url-shortener] [io-18081-exec-4] o.a.c.c.C.[.[.[/].[dispatcherServlet]
: Servlet.service() for servlet [dispatcherServlet] in context with path [] threw exception [Request processing
failed: java.lang.IllegalArgumentException: Illegal character in path at index 7: Invalid code] with root cause

java.net.URISyntaxException: Illegal character in path at index 7: Invalid code
        at java.base/java.net.URI$Parser.fail(URI.java:2995) ~[na:na]
        at java.base/java.net.URI$Parser.checkChars(URI.java:3166) ~[na:na]
        at java.base/java.net.URI$Parser.parseHierarchical(URI.java:3248) ~[na:na]
        at java.base/java.net.URI$Parser.parse(URI.java:3207) ~[na:na]
        at java.base/java.net.URI.<init>(URI.java:645) ~[na:na]
        at java.base/java.net.URI.create(URI.java:930) ~[na:na]
        at com.malinda.url_shortener.controller.UrlController.redirectToOriginalUrl(UrlController.java:42) ~
[!/:0.0.1-SNAPSHOT]
        at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:103) ~
[na:na]
        at java.base/java.lang.reflect.Method.invoke(Method.java:580) ~[na:na]
        ... (50 "at" frames in total)
```

Note on X2. My first attempt passed `é` as a command-line argument, and Windows sent it to curl.exe as a Latin-1 byte. The server correctly rejected that body as invalid UTF-8 JSON (400). That was a client-side artifact, so I resent the URL as a UTF-8 body file; the transcript above is from the resend.

7.2 Check 6a: `--app.base-url=http://localhost:18082` (no trailing slash)

```
== App started: port 18082, --app.base-url=http://localhost:18082 (pid 2050)

### 6. Create with this base-url
$ curl -i -X POST http://localhost:18082/api/urls -H 'Content-Type: application/json' -d
'{"url":"https://baseurl.example.com/page"}'
HTTP/1.1 201
Content-Type: application/json

{"code":"EOQGkA","shortUrl":"http://localhost:18082EOQGkA"}

[server log: no new lines / no stack trace]
```

```
### 6b. Follow it (code EOQGkA)
$ curl -i http://localhost:18082/EOQGkA
HTTP/1.1 302
Location: https://baseurl.example.com/page
Content-Length: 0


[server log: no stack trace]

== App stopped
```

Result: wrong. `shortUrl` is `http://localhost:18082EOQGkA`, with no `/` between the port and the code. The redirect itself works, because only the `shortUrl` string is affected.

7.3 Check 6b: `--app.base-url=http://localhost:18083/` (trailing slash)

```
== App started: port 18083, --app.base-url=http://localhost:18083/ (pid 2089)

### 6. Create with this base-url
$ curl -i -X POST http://localhost:18083/api/urls -H 'Content-Type: application/json' -d
'{"url":"https://baseurl.example.com/page"}'
HTTP/1.1 201
Content-Type: application/json

{"code":"tvCXCG","shortUrl":"http://localhost:18083/tvCXCG"}

[server log: no new lines / no stack trace]

### 6b. Follow it (code tvCXCG)
$ curl -i http://localhost:18083/tvCXCG
HTTP/1.1 302
Location: https://baseurl.example.com/page
Content-Length: 0


[server log: no stack trace]

== App stopped
```

Result: correct. `shortUrl` is `http://localhost:18083/tvCXCG`.

7.4 Stack traces for 4xx cases

Case	Status	Stack trace in server log?
5a empty body	400	No. One `WARN` line from `DefaultHandlerExceptionResolver` ("Required request body is missing")
5b `{}`	400	No, nothing logged
5c `""`	400	No, nothing logged
5d `"  "`	400	No, nothing logged
5e `not-a-url`	400	No, nothing logged
5f `https://`	400	No, nothing logged
5g `javascript:alert(1)`	400	No, nothing logged
5h `ftp://...`	400	No, nothing logged
5i `.../a b`	400	No, nothing logged
X1 malformed JSON (extra)	400	No. One `WARN` line ("JSON parse error")
4 unknown code (should be 404)	**500**	**Yes**. `ERROR` from `dispatcherServlet` plus a `java.net.URISyntaxException` trace (50 frames)

7.5 Extra: `GET /` (no code), default config on port 18084

```
$ curl -i http://localhost:18084/
HTTP/1.1 404
Content-Type: application/json

{"timestamp":"2026-11-08T18:32:00.632Z","status":404,"error":"Not Found","path":"/"}

[server log stack frames: 0]
```

8. Review findings by severity

Order: correctness, then concurrency/data integrity, then performance, then design, then style. Each finding is tagged **[observed]** (reproduced by a test or curl) or **[code reading]**.

8.1 Correctness

C1. Unknown code returns 500 with a stack trace instead of 404. [observed]

- *Where:* `UrlServiceIMPL.redirectToOriginalUrl` (lines 54-60) and `UrlController.redirectToOriginalUrl` (lines 38-43).
- *What:* For an unknown code the service returns the literal string `"Invalid code"`. The controller passes it to `URI.create(...)` (line 42), which throws `IllegalArgumentException` because of the space. Nothing maps that exception, so Spring returns 500 and logs an `ERROR` with a full stack trace.
- *Why it matters:* This breaks the contract (`unknown -> 404`), causes the only failing test, and makes a routine client error look like a server fault in logs and monitoring. It also means a missing record is signalled through the data channel (see D1).
- *Fix direction:* Have the lookup report "not found" explicitly, through the return type or a dedicated exception, instead of a sentinel string. Map that case to 404 in `GlobalExceptionHandler`, so the controller only builds a redirect for a real URL.

C2. `shortUrl` is malformed when `app.base-url` has no trailing slash. [observed]

- *Where:* `UrlServiceIMPL.createShortUrl`, lines 37 and 48 (`baseUrl + existingCode`, `baseUrl + code`).
- *What:* The short URL is built by raw string concatenation. With `--app.base-url=http://localhost:18082` the response contained `"shortUrl":"http://localhost:18082EOQGkA"`. The default config only works because `application.yaml` line 9 now ends in `/`.
- *Why it matters:* A configuration detail that is easy to get wrong silently produces dead links. Nothing fails at startup and no test catches it (see 6.4).
- *Fix direction:* Join the base URL and the code in one place, in a way that is correct whether or not the configured value ends in `/` (normalise once when the value is loaded, or use a URI builder).

C3. A non-ASCII URL comes back changed in `Location`. [observed; mechanism by code reading]

- *Where:* `UrlController.redirectToOriginalUrl`, line 42 (`.location(URI.create(response))`).
- *What:* `https://probe.example.com/café` was accepted (201), but the redirect returned `Location: https://probe.example.com/caf%C3%A9`. Spring writes the `Location` header from the `URI` in its ASCII form, which percent-encodes non-ASCII characters.
- *Why it matters:* The contract says "Location: the original URL, unchanged". Browsers treat the two forms as equivalent, and HTTP headers are ASCII-only anyway, so the risk is low. But the API currently accepts input that it cannot return unchanged.
- *Fix direction:* Decide the contract for non-ASCII input: reject it at validation, or document that it is stored or returned in ASCII-encoded form. Then add a test that pins whichever behaviour you choose.

8.2 Concurrency / data integrity

No findings outside the parked items. `ConcurrentHashMap` makes individual map operations thread-safe. The compound read-then-write in `createShortUrl` and possible code collisions are parked (section 10).

8.3 Performance

No findings outside the parked items. The duplicate lookup's linear scan is parked (section 10).

8.4 Design

D1. The service builds the API response DTO and the short-URL string. [code reading]

- *Where:* `UrlServiceIMPL.createShortUrl` (lines 34-50) and `CreateUrlResult` (wraps `dto.CreateUrlResponse`).
- *What:* The service returns `CreateUrlResult(CreateUrlResponse, boolean)`, so the HTTP response shape and the public-URL formatting (`baseUrl + code`) live in the service layer.
- *Why it matters:* A change to the response JSON or to the public host forces a service change, and the service cannot be reused or tested without the web-facing DTO. It is also why `baseUrl` has to be injected into the service at all.
- *Fix direction:* Have the service return a domain result (the code and whether it was newly created). Let the web layer turn it into `CreateUrlResponse` and build `shortUrl`.

D2. `baseUrl` uses field injection. [code reading]

- *Where:* `UrlServiceIMPL`, lines 21-22 (`@Value("${app.base-url}") private String baseUrl;`).
- *What and why:* The field cannot be `final`, its dependency is hidden from the constructor, and a plain unit test cannot set it without reflection or a Spring context. The controller already uses constructor injection correctly, so the two classes are inconsistent.
- *Fix direction:* Inject the value through the constructor (or a typed configuration-properties object), in line with D1.

D3. Two different error-response shapes. [observed]

- *Where:* `GlobalExceptionHandler` (only handles `InvalidUrlException`).
- *What:* Validation errors return `{"message":"..."}`. An empty body or malformed JSON returns Spring's default `{"timestamp","status","error","path"}` (checks 5a and X1).
- *Why it matters:* Clients must parse two different 400 bodies from the same endpoint.
- *Fix direction:* Choose one error format and make the advice produce it for request-body read failures too (Spring's `ProblemDetail` support is one option).

D4. The service method name `redirectToOriginalUrl` is misleading. [code reading]

- *Where:* `UrlService` line 9 and `UrlServiceIMPL` line 54. The controller method uses the same name at line 38.
- *What and why:* The service does not redirect; it looks up a stored URL. The name hides the method's real job, and that made the sentinel-string design in C1 look natural.
- *Fix direction:* Name the service method for what it does (a lookup or resolve). The controller handler can keep a redirect-oriented name.

D5. The empty-code check can never be true. [code reading + observed]

- *Where:* `UrlServiceIMPL.redirectToOriginalUrl`, line 55 (`code.isEmpty() ||`).
- *What and why:* `GET /{code}` never matches an empty segment (`GET /` returns 404 without reaching the controller; observed in 7.5), so this branch is dead code that implies a case which cannot happen.

- *Fix direction:* Remove the dead condition when reworking the not-found path in C1.

D6. `--enable-preview` in `pom.xml` is unnecessary and not written correctly. [observed + code reading]

- *Where:* `pom.xml`, lines 51-59.
- *What:* No Java preview features are used. `<compilerArgs>` is given as a bare string instead of `<arg>` entries, and the build log shows javac ran with `[debug parameters release 21]`, with no preview flag visible. The explicit `<source>21</source>` and `<target>21</target>` duplicate `<java.version>`, which the Boot parent already applies.
- *Why it matters:* If the flag ever does take effect, compiled classes need `--enable-preview` at runtime and in tests. If it doesn't, the block only misleads readers.
- *Fix direction:* Remove the compiler-plugin override and rely on `<java.version>`.

8.5 Style

- **S1. Unused imports.** [code reading] `UrlServiceIMPL` lines 9-10 (`HttpStatus`, `ResponseEntity`) and `UrlService` line 3 (`CreateUrlResponse`). These are leftovers that make it look as though the service depends on web types.
- **S2. Wildcard import.** [code reading] `UrlController` line 9 (`org.springframework.web.bind.annotation.*`). It hides which annotations the class uses. Use explicit imports.
- **S3. Double spaces and spacing.** [code reading] `UrlController` line 38 (`public  ResponseEntity`), line 40 (`return  ResponseEntity`) and line 26 (`created()){`). `InvalidUrlException` line 3 (`RuntimeException{`). `UrlServiceIMPL` lines 27-28 (two blank lines). Running the IDE formatter would fix all of these.
- **S4. All-caps `IMPL` naming.** [code reading] Package `service.IMPL` and class `UrlServiceIMPL`. Java convention is lowercase package names and CamelCase class names (for example `service.impl` and `UrlServiceImpl`, or a name that describes the implementation, such as an in-memory one).
- **S5. Magic number 6.** [code reading] `ShortCodeGenerator.generateShortCode`, line 14. The code length is a domain rule and should be a named constant.
- **S6. Naming nits.** [code reading] `UrlValidator` line 16 names the variable `schema`; the URI concept is "scheme". `UrlController` names both the `CreateUrlResult` (line 24) and the looked-up URL string (line 39) `response`, although neither is a response. The 201/200 branch at lines 26-34 repeats the `.body(response.response())` chain where only the status differs.

9. What's done well

- **Validation is in one place** (`UrlValidator`). It has a dedicated `InvalidUrlException`, mapped to 400 by a `@RestControllerAdvice`. All 8 invalid inputs and the empty body return 400 with a clear message and **no stack trace**.
- **Duplicate handling matches the contract.** A resubmitted URL returns 200 with the same `code` and `shortUrl` (test #7 and curl check 2), and the `created` flag keeps the 201/200 decision explicit.
- **Redirects are correct for normal URLs.** They return 302, and query strings and percent-encoding are preserved exactly (`?q=hello%20world`, `?utm_source=...`). There is no double-encoding.
- The scheme check is case-insensitive and correctly rejects `javascript:` and `ftp:`.
- **Constructor injection** in `UrlController`, and a service interface the controller depends on.
- **Records** for every DTO, so they are immutable and concise.
- `SecureRandom` for codes and `ConcurrentHashMap` for the store are sensible choices for this stage.
- The endpoint (`/api/urls`), the request field (`url`) and the response fields (`code`, `shortUrl`) now exactly match the contract, and the earlier `generateSHortCode` typo is fixed.
- Commits are small and their messages describe the change.

10. Future-stage flags

- A newly generated code can equal an existing code, and the new mapping silently replaces the old one.
- Two simultaneous requests for the same new URL can both get 201 with different codes.
- Duplicate lookup time grows linearly with the number of stored links.
- Code generation is a static call, so tests cannot control which code is produced.

11. Comparison with the 5 November review

No 5 November review file exists in `docs/reviews/`, so this section uses the list supplied by the author. The evidence covers the two commits since then: `865430c` (2026-11-06) and `cd61f9d` (2026-11-08).

#	5 November item	Status	Evidence
1	Unknown code returned 500	**Open**	Still 500 with a stack trace (curl check 4; test #5 fails). See C1
2	Blank or missing url returned 500	**Fixed**	`{}`, `""` and `"  "` each give 400 `URL cannot be empty` (`UrlValidator` lines 9-11; tests #8-10; curl 5b-5d). An empty body also gives 400 (5a)
3	Duplicate URL returned 500	**Fixed**	`865430c` still threw a bare `RuntimeException` for duplicates. HEAD returns 200 with the existing code (`UrlServiceIMPL` lines 29-41; test #7; curl check 2)
4	URL format not validated	**Fixed**	`UrlValidator` checks the URI syntax, an http(s) scheme and a non-null host. All 8 invalid inputs give 400 (tests #8-15; curl 5b-5i). Edge case: non-ASCII input is accepted but returned encoded (C3)
5	An empty-code check that can never be true	**Open**	`UrlServiceIMPL` line 55 `code.isEmpty() ||` is unchanged (D5)
6	The lookup returned an error string as data	**Open**	`UrlServiceIMPL` line 56 `return "Invalid code";` is unchanged and causes C1
7	Bare RuntimeExceptions with nothing mapping them to statuses	**Partly**	The bare `throw new RuntimeException()` calls from `865430c` are gone. `InvalidUrlException` and `GlobalExceptionHandler` now map validation errors to 400. Not-found has no exception or mapping, and the `IllegalArgumentException` from `URI.create` reaches the container as a 500
8	shortUrl depended on a trailing slash in base-url	**Open**	The concatenation is unchanged (`UrlServiceIMPL` lines 37, 48). `865430c` added `/` to the yaml value, which hides the problem. Without the slash the response was `http://localhost:18082EOQGkA` (7.2). See C2
9	The service returned the API response DTO and built the short-URL string	**Open**	The service still builds `CreateUrlResponse(code, baseUrl + code)`, now wrapped in `CreateUrlResult` (D1)
10	Field injection for baseUrl	**Open**	`UrlServiceIMPL` lines 21-22 are unchanged (D2)
11	Misleading method name redirectToOriginalUrl	**Open**	`UrlService` line 9 and `UrlServiceIMPL` line 54 are unchanged (D4)
12	--enable-preview in the pom	**Open**	`pom.xml` line 57 is unchanged; the pom has not changed since 2026-11-02 (D6)
13	Unused imports	**Open**	`UrlServiceIMPL` lines 9-10 and `UrlService` line 3 (S1). They were added in `cd61f9d`
14a	All-caps IMPL naming	**Open**	`service/IMPL/UrlServiceIMPL.java` (S4)

#	5 November item	Status	Evidence
14b	Double spaces	**Open**	`UrlController` lines 38 and 40 (S3)
14c	Wildcard import	**Open**	`UrlController` line 9 (S2)
14d	Magic number 6	**Open**	`ShortCodeGenerator` line 14 (S5)
15	No tests for duplicates or invalid input	**Partly**	These cases are now covered, but only by `UrlShortenerErrorCasesTest`, which the reviewer created in this session (Step 2). `git log -- src/test` shows no author-written test changes since `234e362` (2026-11-02)

Tally: **3 fixed, 2 partly, 13 open** (18 rows, counting 14a-14d separately).

12. Facts only

Redirect status. `302 Found` (`HttpStatus.FOUND`, `UrlController` line 41). Observed in curl checks 1b, 3b and 6.

How short codes are generated. `ShortCodeGenerator.generateShortCode()` is a static method. It loops 6 times, and on each pass picks `CHARACTERS.charAt(secureRandom.nextInt(62))` from `0-9a-zA-Z` (62 characters) using one shared static `SecureRandom`. The result is 6 characters, giving 62^6 (about 5.68 x 10^10) possible codes. No uniqueness check is made before the code is stored.

How the existing code is found for a duplicate. `UrlServiceIMPL.createShortUrl` first validates the input. It then iterates over `urlMappings.entrySet()` (a `ConcurrentHashMap<String code, String url>`) and compares each stored value with the raw input using `String.equals`. The input is not normalised (no trimming, case folding or trailing-slash handling), so `.../42/?ref=manual` and `.../42?ref=manual` get different codes (curl X3). The key of the first matching entry is returned as the existing code.

How the controller decides between 201 and 200. The service returns `CreateUrlResult(CreateUrlResponse response, boolean created)`. `created` is `false` when the scan found an existing entry and `true` when a new code was generated and stored. The controller returns `201 CREATED` when `response.created()` is true and `200 OK` otherwise, with the same `CreateUrlResponse` body in both cases.

How each failure maps to an HTTP status.

Failure	Raised where	Handled by	Status / body
`url` missing (`{}`), `null`, empty or whitespace-only	`UrlValidator` line 10, `InvalidUrlException("URL cannot be empty")`	`GlobalExceptionHandler`	400 `{"message":"URL cannot be empty"}`
`URI.create` throws (illegal characters such as a space, or `https://` with no authority)	`UrlValidator` line 22, `InvalidUrlException("Malformed URL")`	`GlobalExceptionHandler`	400 `{"message":"Malformed URL"}`
No scheme, a scheme other than http/https, or no host	`UrlValidator` line 19, `InvalidUrlException("Invalid URL format")`	`GlobalExceptionHandler`	400 `{"message":"Invalid URL format"}`

Failure	Raised where	Handled by	Status / body
Empty request body or malformed JSON	Spring `HttpMessageNotReadableException`	Spring `DefaultHandlerExceptionResolver` (one WARN line)	400 Spring default body `{"timestamp","status","error","path"}`
Unknown code	Service returns `"Invalid code"`; `URI.create` throws `IllegalArgumentException` in the controller	Nothing (unhandled)	**500** Spring default body, plus an ERROR log with a stack trace
`GET /` (no code)	No handler matches	Spring	404 Spring default body, no stack trace (section 7.5)

The exact URL validation rules (`UrlValidator.validateUrl`, applied in order to the raw string):

1. Reject with "URL cannot be empty" if the value is `null` or `isBlank()` (empty, or whitespace only).
2. Parse with `java.net.URI.create(value)`. If it throws `IllegalArgumentException` (any RFC 2396 syntax error as implemented by `java.net.URI`, including unencoded spaces, leading or trailing whitespace, and `https://` with no authority), reject with "Malformed URL".
3. Reject with "Invalid URL format" if `getScheme()` is `null`, or is neither `http` nor `https` (compared case-insensitively).
4. Reject with "Invalid URL format" if `getHost()` is `null`. `java.net.URI` returns `null` for a missing host and for an authority it cannot parse as a server-based host.
5. Nothing else is checked: there are no length, port-range, userinfo, TLD or non-ASCII rules. The original string, not a normalised form, is what gets stored.

Authorship

- **This session:** I did **not** write, edit, reformat or delete any file under `src/main`, `pom.xml`, `application.yaml` or any other config. `git status --short` at the end of testing showed one untracked file, `src/test/java/com/malinda/url_shortener/UrlShortenerErrorCasesTest.java`, which I created as Step 2 allowed. I also created `docs/reviews/stage-1-final-review.md` and `docs/reviews/stage-1-final-review.pdf`. Build outputs went to `target/`, and scratch scripts and logs went to a temporary directory outside the repo. I did not commit or push.
- **Earlier sessions:** I can't see earlier sessions, so I can't say whether Claude wrote or edited any `src/main` code in them. The repository's `CLAUDE.md` (added in `cd61f9d`) tells Claude never to create or edit files under `src/main`. All commits are authored by `malinda-sampath`.